

Manual

Upgrade: Detailed Planning Data for ERP EIS-EFD 8.2

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1 Upgrade: Detailed Planning Data for ERP

Possible fields of application

The function package "modification of detailed scheduling data to ERP" enhances the ERP systems interface by the upload of detailed scheduling information.

Implementation notes

The function package is used if you:

- use the detailed scheduling/shop floor scheduling module and
- would like to upload the detailed scheduling dates specified there to the ERP system

Integration

The function package uses the data specified within the framework of detailed scheduling.

Functions

- Upload of changes to the planning
 - Upload of detailed scheduling dates
 - Upload of the machine/workplace on which the OP was scheduled
 - Identification of conflicts

2 Setup of Data Record Structure

The data are transferred in the following structure. Within this structure the value of the SEGNAM field precisely defines the set-up of the user data structure in the SDATA field.

Field name	Type	Length	Designation	Data field and meaning
SEGNAM*	Char	30	Segment name	This field is occupied by the writing system with the respective segment name. This precisely defines the set-up of the data record (SDATA field). Example: HY72_AU_HD_001
MANDT*	Char	3	Client	Reserved; fixed: '000'
DOCNUM*	Char	16	IDOC number	Serial number for the IDOCs Reserved: fixed '0000000000000000'
SEGNUM*	Char	6	Segment number	Reserved: fixed '000000'
PSEGNUM	Char	6	Parent segment number	Reserved; fixed: '000000'
HLEVEL	Char	2	Hierarchy level	Reserved; fixed: '00'
SDATA	Char	1000	User data	This field contains the user data. The structure of this field is defined by the SEGNAM field.

3 Uploading data types: HYDRA --> ERP

Type	Description
CHAR x	Information is left-aligned for the data type CHAR; unnecessary places are filled with blanks. Example: "ABCD "
NUM x	Numeric field of the length x without sign. Numbers are right-aligned; unnecessary places are filled with zeros. Example: "00000002"
DEC_O x.y	Numeric field of the length x and y decimal places. An algebraic sign is preceding the data field ("+" or "-"). Places that are not required are filled with zeros. There is NO DECIMAL SEPARATOR. e.g. DEC_O 13,3: • -1234567890,123 → -1234567890123 • 234567890,3 → +0234567890300
DATE	The date is displayed in the YYYYMMDD format. The field is filled with blanks (if it is not required).
TIME	The time is transferred in the HHMMSS format. The field is populated with "000000".

Generally, HYDRA always transfers a contiguous data structure. Data fields that are not used are filled with blanks. The following definitions apply if you use the file port:

Each data record included in the file has to be completed by 'CR' (U+000D) and 'LF' (U+000A) for Windows and 'LF' (U+000A) for Unix.

HYDRA expects the file to be in the UTF-8 format and HYDRA also uses this format for uploads. On request, the file transfer may also be performed in the file format that was used until MW 2.0.

4 Upload of Planning Changes

PPS specifications can be modified or infringed by assigning operations in the HYDRA shop floor scheduling module (HYDRA-HLS). These modifications can be uploaded to the PPS system.



Only the modifications in the graphical planning board of the HYDRA shop floor scheduling system (HLS) will be uploaded. Modifications made by the editing functions for operations or the order sequencing will not be taken into account.

The following modifications and/ or statuses will be confirmed/uploaded:

- Planning for a workplace in HLS
- Re-planning within a workplace in HLS
- Re-planning for a different workplace (workplace/ machine change) in HLS
- Deallocation of the operation in HLS

The customer will interpret the confirmations/uploads as well as the processing resulting for the PPS system.

The following optional parameters can be set at the order type using an upload indicator (HYDRA customizing) for MLE communication:

- Upload of all operations to the PPS system .
(Default: only the operations generated by PPS will be uploaded)
- Only upload of delayed operations
(default: inactive)
- Only upload of operations with different machine occupation/assignment
(default: inactive)

The below specifications apply for the Idoc:

Message type: HY72ADRCK_SC

IDOC type: HY72ADRCK_SC

Segment file name: HY72ADRCK_SCHEDULE

File extension: According to the configuration in the MLE communication
(Logical systems > Outbound Configuration Fileport)
Normally: ".dat"

Field	T	L	D	Description
ANR	CHAR	40		HYDRA order number = combined order/ operation number The precise length that will be confirmed/uploaded depends on the length configuration of the order and/or operation in the HYDRA basic settings.
RMNR	CHAR	40		Upload number (if available in HYDRA)
AUART	CHAR	5		Order type of an order; according to HYDRA configuration
DAT	DATE	8		Date of fixing (saving) of the reallocation in HLS
ZEI	TIME	6		Time of fixing (saving) of the reallocation in HLS
SGR:GUTB	DEC_O	13	3	Target quantities in base quantity unit if stored to the operation.
SGE:B	CHAR	3		Base quantity unit if stored to the operation.
SGR:GUTP	DEC_O	13	3	Target quantities in primary quantity unit
SGE:P	CHAR	3		Primary quantity unit: Primary unit of entry from the operation
DATB	DATE	8		These data fields include the start time scheduled by HLS for this operation.
ZEIB	TIME	6		These fields are not defined with the ACTION = "G".
DATE	DATE	8		These data fields include the end time scheduled by HLS for this operation.
ZEIE	TIME	6		These fields are not defined with the ACTION = "G".
MNR	CHAR	8		Workplace, to which the operation was allocated. This field is not defined with the ACTION = "G".
MGRP	CHAR	8		ACTION = "M" or "U": Group of the workplace, to which the operation was allocated. ACTION = "G": Group, to which the operation was deallocated.
ART	CHAR	1		"K" = if a conflict is found when planning changes are uploaded for the operation (see KONFLIKT field), otherwise undefined.
AKTION	CHAR	2		"M " - operation was allocated to workplace MNR. This may be an allocation from the pool of groups to a workplace as well as a reallocation of one workplace to another. "U " - Operation was rescheduled within the workplace MNR. "G " - Operation was deallocated to the pool of groups.
KONFLIKT	CHAR	2		Prio 1: "PE" - The end date of the operation exceeds the basic date of the order specified by the PPS system Prio 2: "TE" - The operation exceeds the latest end date resulting from scheduling. Prio 3: "PM" - Workplace is not the workplace transferred by the PPS system (during the first data transfer of the operation). Only possible if the operation was transferred by the PPS system. PLEASE NOTE: Only the conflict of the highest priority will be uploaded. Example: If a planned operation has exceeded the basic date of the order and if an operation was allocated to another scheduled workplace transferred by the PPS system, only the "PE" conflict will be uploaded due to scheduling.

If the additional function to split operations is used in the HYDRA shop floor scheduling system (HLS-AGS license),



- split masters will not be uploaded since they are no longer planned after splitting;
- split operations will only be uploaded if the "Upload PPS operations only" option is NOT set in the order type, "confirmations" tab.

5 HYDRA Settings relevant for the Application

Proceed as described in the following to activate the interface for the upload of planning changes from the HYDRA Shop Floor Scheduling (HLS):

Activating the logging

Edit the following entries in the [Logging configuration](#):

Parameter name	Value
For the scheduling of operations	
Object	HLS
Action	EINPLANEN
Logging of dialog data	Yes
Logging	Entire object
Comment	No
Labeling	No
Segment – for the upload to non-SAP systems	HY72ADRCK_SCHEDULE
Segment – for the upload to SAP systems	Z2HY72ADRCK_SCHEDULE000X000
For the replanning of operations	
Object	HLS
Action	UMPLANEN
Logging of dialog data	Yes
Logging	Entire object
Comment	No
Labeling	No
Segment – for the upload to non-SAP systems	HY72ADRCK_SCHEDULE
Segment – for the upload to SAP systems	Z2HY72ADRCK_SCHEDULE000X000
For the deallocation of operations	
Object	HLS
Action	AUSPLANEN
Logging of dialog data	Yes
Logging	Entire object
Comment	No
Labeling	No
Segment – for the upload to non-SAP systems	HY72ADRCK_SCHEDULE
Segment – for the upload to SAP systems	Z2HY72ADRCK_SCHEDULE000X000

Activating the upload for the order type

If required, activate for the relevant [Order type](#) (s) > Confirmation > Confirmation of planning changes in shop floor scheduling:

Parameter name	Value
Upload PPS operations only	Activate if required
Upload delayed operations only	Activate if required
Upload operations with deviating assignment only	Activate if required

Upload PPS operations only

Upload PPS operations only

If you activate this option, the operations created in HYDRA are not uploaded.

Upload delayed operations only

Upload delayed operations only

If you activate this option, only delayed operations are uploaded.

Upload operations with deviating assignment only

Upload operations with deviating assignment only

If you activate this option, only operations are uploaded that have been planned for another workplace than the workplace initially transferred.

You can use the option *Upload PPS operations only* no matter whether the other options are enabled or not. But you can only enable one of the two options *Upload delayed operations only* and *Upload operations with deviating assignment only*.

Activating the upload in the scheduler

Use the [HYDRA Scheduler](#) to plan jobs for the outbound processing:

Parameter name	Value
Transfer of uploads to the MLE outbound transactions	
Product key	
License key	
Command (Windows):	sh.exe ./myerprck.scr /LOGGING /LOGGING_SEGNAM=HY72ADRCK_SCHEDULE
Command (Unix):	./myerprck.scr /LOGGING /LOGGING_SEGNAM=HY72ADRCK_SCHEDULE
Comment:	EIS-EFD: Upload of planning changes -> MLE outbound transactions

Parameter name	Value
Interval	5
Upload of confirmations from the MLE outbound transactions to the ERP system	
Product key	
License key	
Command (Windows) for the upload to a non-SAP system:	sh.exe ./hysapupl.scr /UPLSEGNAM=HY72ADRCK_SCHEDULE
Command (Windows) for the upload to an SAP system:	sh.exe ./hysapupl.scr /UPLSEGNAM=Z2HY72ADRCK_SCHEDULE000X000
Command (Unix) for the upload to a non-SAP system:	./hysapupl.scr /UPLSEGNAM=HY72ADRCK_SCHEDULE
Command (Unix) for the upload to an SAP system:	./hysapupl.scr /UPLSEGNAM=Z2HY72ADRCK_SCHEDULE000X000
Comment:	EIS-EFD: Upload of planning changes MLE outbound transactions -> ERP system
Interval	5

Editing the HYDRA distribution model – output for non-SAP systems

Use the [HYDRA distribution model](#) to edit an entry for the HYDRA outbound processing:

Parameter name	Value
To upload time tickets	
Message type	HY72ADRCK_SC
Description	EIS-EFD – upload of planning changes
IDoc type	HY72ADRCK_SC
Retention period	10
Log. target system	Created logical system
Segment name 1	HY72ADRCK_SCHEDULE

Editing the HYDRA distribution model – output for SAP systems

Use the [HYDRA distribution model](#) to edit an entry for the HYDRA outbound processing:

Parameter name	Value
To upload time tickets	
Message type	ZHY72ADRCK_SC
Description	EIS-EFD – upload of planning changes
IDoc type	ZHY72ADRCK_SC01
Retention period	10

Parameter name	Value
Log. target system	Created logical system
Segment name 1	Z2HY72ADRCK_SCHEDULE000X000

6 Test Files

Overview

Attached to this documentation, you will find test files for the interface EIS-EFD. The attachment is only available, if the documentation is in PDF format.

The documentation [Open PDF attachments](#) describes how to call the attached test files.

The following test files are attached to the PDF document:

File	Type	Comment
HY72ADRCK_SCHEDULE.dat	Outbound processing	Sample file for the upload of planning changes in HYDRA standard format